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INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B. Tech. M.Tech./ M. Tech. (CSE)

Semester: VIII

SUBJECT CODE: MCS202

PARALLEL COMPUTING

Time: 03:00 Hrs. **Maximum Marks: 70**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. **(14 Marks)**

Q. 1. Answer the following questions: **(1×7 = 7 Marks)**

(i) Give one characteristics of parallel computing.
(ii) SIMD stands for?
(iii) Explain bitonic sort.
(iv) Write difference between shared and distributed memory.
(v) What is cache memory?
(vi) Give one application of parallel computing.
(vii) What is Data parallelism?

Q. 2. Answer the following questions: **(1×7 = 7 Marks)**

(i) Define Flynn's classification.
(ii) Define Amdahl's Law.
(iii) Give one difference between multiprogramming and multiprocessing.
(iv) Explain ring topology.
(v) Name any two parallel programming models.
(vi) What is the purpose of load balancing in parallel computing?
(vii) What is the difference between task parallelism and data parallelism?

SECTION – B

NOTE:Attempt all questions. **(7×2×2 = 28 Marks)**

Q.3.(i)(a) Explain shared memory and distributed memory architectures in parallel computing. Compare their advantages and disadvantages.

OR

(b) Describe the concept of message passing in parallel computing. Explain how MPI (Message Passing Interface) works.

(ii)(a) What are the major applications of parallel computing? Discuss its use in AI, scientific simulations, and big data analytics.

OR

(b) Explain the Bitonic Sort algorithm with an example. How does it work in a parallel computing environment?

Q.4.(i)(a) Explain the PRAM model and its types (EREW, CREW, CRCW).

OR

(b) Explain Amdahl's Law with suitable example.

(ii)(a) Write the algorithm for List Ranking and explain with example.

OR

(b) Write the algorithm for odd even transposition. Also sort the sequence A, H, D, C, G, E, B, G by applying odd even transposition method.

SECTION – C

NOTE: Attempt all questions. **(14×2 = 28 Marks)**

Q. 5. (a) How can load balancing be achieved in parallel sorting algorithms? Explain the techniques used to distribute tasks efficiently among processors.

OR

(b) Explain the different types of parallelism (Task Parallelism, Data Parallelism, and Instruction-Level Parallelism) with suitable examples.

Q. 6. (a) Compare and contrast parallel computing and distributed computing. Discuss their differences, advantages, and use cases in modern computing.

OR

(b) Explain the concept of cache coherence in parallel computing. Discuss the challenges it presents in a multiprocessor system and explain different cache coherence protocols such as MESI, MOESI, and MSI.